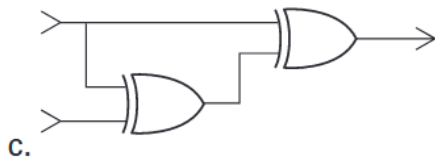
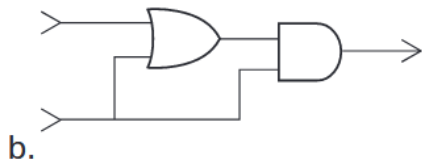
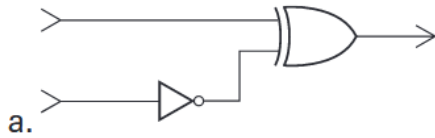


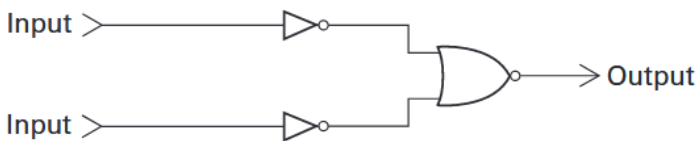
Chapter Review Problems

(Asterisked problems are associated with optional sections.)

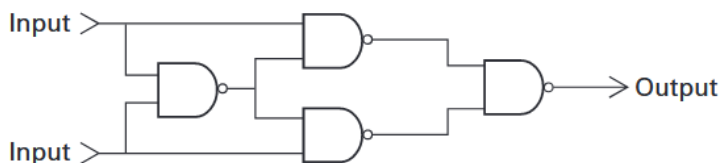
- Determine the output of each of the following circuits, assuming that the upper input is 1 and the lower input is 0. What would be the output when upper input is 0 and the lower input is 1?



- a. What Boolean operation does the circuit compute?

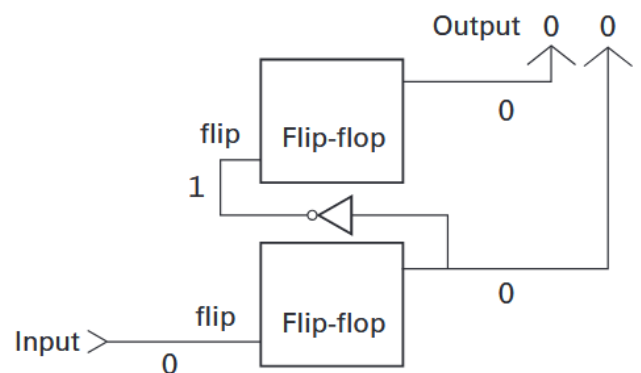


- b. What Boolean operation does the circuit compute?

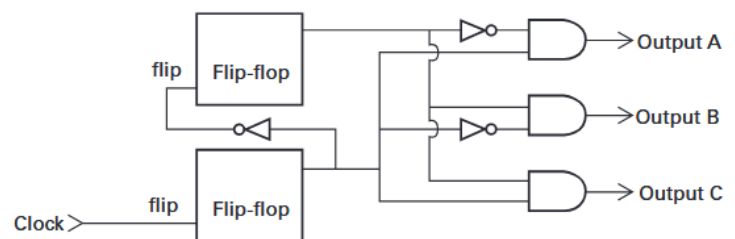


- a. If we were to purchase a flip-flop circuit from an electronic component store, we may find that it has an additional input called *flip*. When this input changes from a 0 to 1, the output flips state (if it was 0 it is now 1 and vice versa). However, when the flip input changes from 1 to a 0, nothing happens. Even though we may not know the details of the circuitry needed to accomplish this behavior, we could still

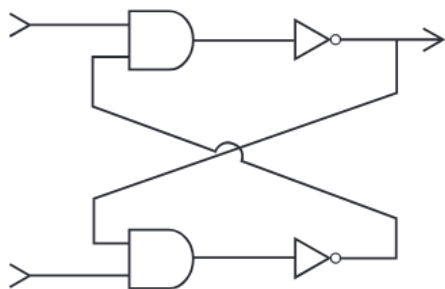
use this device as an abstract tool in other circuits. Consider the circuitry using two of the following flip-flops. If a pulse were sent on the circuit's input, the bottom flip-flop would change state. However, the second flip-flop would not change, since its input (received from the output of the NOT gate) went from a 1 to a 0. As a result, this circuit would now produce the outputs 0 and 1. A second pulse would flip the state of both flip-flops, producing an output of 1 and 0. What would be the output after a third pulse? After a fourth pulse?



- b. It is often necessary to coordinate activities of various components within a computer. This is accomplished by connecting a pulsating signal (called a clock) to circuitry similar to part a. Additional gates (as shown) send signals in a coordinated fashion to other connected circuits. On studying this circuit, you should be able to confirm that on the 1st, 5th, 9th . . . pulses of the clock, a 1 will be sent on output A. On what pulses of the clock will a 1 be sent on output B? On what pulses of the clock will a 1 be sent on output C? On which output is a 1 sent on the 4th pulse of the clock?



4. Assume that both of the inputs in the following circuit are 1. Describe what would happen if the upper input were temporarily changed to 0. Describe what would happen if the lower input were temporarily changed to 0. Redraw the circuit using NAND gates.



5. The following table represents the addresses and contents (using hexadecimal notation) of some cells in a machine's main memory. Starting with this memory arrangement, follow the sequence of instructions and record the final contents of each of these memory cells:

Address	Contents
00	AB
01	53
02	D6
03	02

Step 1. Move the contents of the cell whose address is 03 to the cell at address 00.

Step 2. Move the value 01 into the cell at address 02.

Step 3. Move the value stored at address 01 into the cell at address 03.

6. How many cells can be in a computer's main memory if each cell's address can be represented by two hexadecimal digits? What if four hexadecimal digits are used?
7. What bit patterns are represented by the following hexadecimal notations?
- 8A9
 - DCB
 - EF3
 - A01
 - C99
8. What is the value of the least significant bit in the bit patterns represented by the following hexadecimal notations?
- 9A
 - 90
 - 1B
 - 6E

9. Express the following bit patterns in hexadecimal notation:
- 10110100101101001011
 - 000111100001
 - 111111011011011
10. Suppose a digital camera has a storage capacity of 500MB. How many black-and-white photographs could be stored in the camera if each consisted of 512 pixels per row and 512 pixels per column if each pixel required one bit of storage?
11. Suppose an image is represented on a display screen by a square array containing 256 columns and 256 rows of pixels. If for each pixel, 3 bytes are required to encode the color and 8 bits to encode the intensity, how many byte-size memory cells are required to hold the entire picture?
12. a. What are the advantages, if any, of using zoned-bit recording?
b. What is the difference between seek time and access time?
13. Suppose that you want to create a backup of your entire data which is around 10GB. Would it be reasonable to use DVDs for the purpose of creating this backup? What about BDs (Blu-ray Disks)?
14. If each sector on a magnetic disk can store 512 bytes of data, how many sectors are required to store two pages of integers (perhaps 10 lines of 100 integers in each page) if every integer is represented by using four bytes?
15. How many bytes of storage space would be required to store a 20-page document containing details of employees, in which each page contains 100 records and every record is of 200 characters, if two byte Unicode characters were used?
16. In zoned-bit recording, why does the rate of data transfer vary, depending on the portion of the disk being used?
17. What is the average access time for a hard disk which has a rotation delay of 10 milliseconds and a seek time of 9 milliseconds?
18. Suppose a disk storage system consists of 5 platters with 10 tracks on each side and

8 sectors in each track. What is the capacity of the system? Assume every sector contains 512 bytes and data can be stored on both surfaces of each platter.

19. Here is a message in ASCII. What does it say?
01000011 01101111 01101101 01110000
01110101 01110100 01100101 01110010
00100000 01010011 01100011 01101001
01100101 01101110 01100011 01100101
00100001
20. The following two messages are encoded in ASCII using one byte per character and then represented in hexadecimal notation. Are both the messages same?
436F6D7075746572 436F6D7075736572
21. Encode the following sentences in ASCII using one byte per character.
a. Is 1 byte = 8 bits?
b. Yes, a byte contains 8 bits!
22. Combine the two sentences of the previous problem and express it in hexadecimal notation.
23. List the hexadecimal representations of the integers from 20 to 27.
24. a. Write the number 100 by representing 1 and 0 in ASCII.
b. Write the number 255 in binary representation.
25. What values have binary representations in which only one of the bits is 1? List the binary representations for the smallest six values with this property.
- *26. Convert each of the following hexadecimal representations to binary representation and then to its equivalent base 10 representation:
a. A b. 14 c. 1E
d. 28 e. 32 f. 3C
g. 46 h. 65 i. CA
j. 12F k. 194 l. 1F9
- *27. Convert each of the following base 10 representations to its equivalent binary representation:
a. 110 b. 99 c. 72
d. 81 e. 36
- *28. Convert each of the following excess 32 representations to its equivalent base 10 representation:
a. 011111 b. 100110 c. 111000
d. 000101 e. 010101
- *29. Convert each of the following base 10 representations to its equivalent excess sixteen representation:
a. -12 b. 0 c. 10
d. -8 e. 9
- *30. Convert each of the following two's complement representations to its equivalent base 10 representation:
a. 010101 b. 101010 c. 110110
d. 011011 e. 111001
- *31. Convert each of the following base 10 representations to its equivalent two's complement representation in which each value is represented in 8 bits:
a. -27 b. 3 c. 21
d. 8 e. -18
- *32. Perform each of the following additions assuming the bit strings represent values in two's complement notation. Identify each case in which the answer is incorrect because of overflow.
a. 00101 + 01000 b. 11111 + 00001
c. 01111 + 00001 d. 10111 + 11010
e. 11111 + 11111 f. 00111 + 01100
- *33. Solve each of the following problems by translating the values into two's complement notation (using patterns of 5 bits), converting any subtraction problem to an equivalent addition problem, and performing that addition. Check your work by converting your answer to base 10 notation. (Watch out for overflow.)
a. 5 + 1 b. 5 - 1 c. 12 - 5
d. 8 - 7 e. 12 + 5 f. 5 - 11
- *34. Convert each of the following binary representations into its equivalent base 10 representation:
a. 11.11 b. 100.0101 c. 0.1101
d. 1.0 e. 10.01
- *35. Express each of the following values in binary notation:
a. $5\frac{3}{4}$ b. $15\frac{15}{16}$ c. $5\frac{3}{8}$
d. $1\frac{1}{4}$ e. $6\frac{5}{8}$
- *36. Decode the following bit patterns using the floating-point format described in Figure 1.24:
a. 01011001 b. 11001000
c. 10101100 d. 00111001

- *37. Encode the following values using the 8-bit floating-point format described in Figure 1.24. Indicate each case in which a truncation error occurs.
- a. $-7\frac{1}{2}$ b. $\frac{1}{2}$ c. $-3\frac{3}{4}$
d. $\frac{7}{32}$ e. $\frac{31}{32}$
- *38. Assuming you are not restricted to using normalized form, list all the bit patterns that could be used to represent the value $3/8$ using the floating-point format described in Figure 1.24.
- *39. What is the best approximation to the square root of 2 that can be expressed in the 8-bit floating-point format described in Figure 1.24? What value is actually obtained if this approximation is squared by a machine using this floating-point format?
- *40. What is the best approximation to the value one-tenth that can be represented using the 8-bit floating-point format described in Figure 1.24?
- *41. Explain how errors can occur when measurements using the metric system are recorded in floating-point notation. For example, what if 110 cm was recorded in units of meters?
- *42. One of the bit patterns 01011 and 11011 represents a value stored in excess 16 notation and the other represents the same value stored in two's complement notation.
- a. What can be determined about this common value?
b. What is the relationship between a pattern representing a value stored in two's complement notation and the pattern representing the same value stored in excess notation when both systems use the same bit pattern length?
- *43. The three bit patterns 10000010, 01101000, and 00000010 are representations of the same value in two's complement, excess, and the 8-bit floating-point format presented in Figure 1.24, but not necessarily in that order. What is the common value, and which pattern is in which notation?
- *44. Which of the following values cannot be represented accurately in the floating-point format introduced in Figure 1.24?
- a. $6\frac{1}{2}$ b. $\frac{13}{16}$ c. 9
d. $\frac{17}{32}$ e. $\frac{15}{16}$
- *45. If you changed the length of the bit strings being used to represent integers in binary from 4 bits to 6 bits, what change would be made in the value of the largest integer you could represent? What if you were using two's complement notation?
- *46. What would be the hexadecimal representation of the largest memory address in a memory consisting of 4MB if each cell had a one-byte capacity?
- *47. What would be the encoded version of the message
xxy yyx xxy xxy yyx
if LZW compression, starting with the dictionary containing x, y, and a space (as described in Section 1.8), were used?
- *48. The following message was compressed using LZW compression with a dictionary whose first, second, and third entries are x, y, and space, respectively. What is the decompressed message?
22123113431213536
- *49. If the message
xxy yyx xxy xxyy
were compressed using LZW with a starting dictionary whose first, second, and third entries were x, y, and space, respectively, what would be the entries in the final dictionary?
- *50. As we will learn in the next chapter, one means of transmitting bits over traditional telephone systems is to convert the bit patterns into sound, transfer the sound over the telephone lines, and then convert the sound back into bit patterns. Such techniques are limited to transfer rates of 57.6 Kbps. Is this sufficient for teleconferencing if the video is compressed using MPEG?
- *51. Encode the following sentences in ASCII using even parity by adding a parity bit at the high-order end of each character code:
- a. Does $100/5 = 20$?
b. The total cost is \$7.25.
- *52. The following message was originally transmitted with odd parity in each short bit string. In which strings have errors definitely occurred?
11001 11011 10110 00000 11111 10001
10101 00100 01110

Chapter Review Problems

(Asterisked problems are associated with optional sections.)

1. a. In what way are general-purpose registers and main memory cells similar?
b. In what way do general-purpose registers and main memory cells differ?
2. Answer the following questions in terms of the machine language described in Appendix C.
 - a. Write the instruction 2304 (hexadecimal) as a string of 16 bits.
 - b. Write the op-code of the instruction B2A5 (hexadecimal) as a string of 4 bits.
 - c. Write the operand field of the instruction B2A5 (hexadecimal) as a string of 12 bits.
3. Suppose a block of data is stored in the memory cells of the machine described in Appendix C from address 98 to A2, inclusive. How many memory cells are in this block? List their addresses.
4. What is the value of the program counter in the machine described in Appendix C immediately after executing the instruction B0CD?
5. Suppose the memory cells at addresses 00 through 05 in the machine described in Appendix C contain the following bit patterns:

Address	Contents
00	22
01	11
02	32
03	02
04	C0
05	00

Assuming that the program counter initially contained 00, record the contents of the program counter, instruction register, and memory cell at address 02 at the end of each fetch phase of the machine cycle until the machine halts.

6. Suppose three values x , y , and z are stored in a machine's memory. Describe the sequence of events (loading registers from memory, saving values in memory, and so on) that leads to the computation of $x + y + z$. How about $(2x) + y$?
7. The following are instructions written in the machine language described in Appendix C. Translate them into English.

- a. 7123 b. 40E1 c. A304
d. B100 e. 2BCD

8. Suppose a machine language is designed with an op-code field of 4 bits. How many different instruction types can the language contain? What if the op-code field is increased to 6 bits?
9. Translate the following instructions from English into the machine language described in Appendix C.
 - a. LOAD register 6 with the hexadecimal value 77.
 - b. LOAD register 7 with the contents of memory cell 77.
 - c. JUMP to the instruction at memory location 24 if the contents of register 0 equals the value in register A.
 - d. ROTATE register 4 three bits to the right.
 - e. AND the contents of registers E and 2 leaving the result in register 1.
10. Rewrite the program in Figure 2.7 assuming that the values to be added are encoded using floating-point notation rather than two's complement notation.
11. Classify each of the following instructions (in the machine language of Appendix C) in terms of whether its execution changes the contents of the memory cell at location 3B, retrieves the contents of the memory cell at location 3C, or is independent of the contents of the memory cell at location 3C.
 - a. 353C b. 253C c. 153C
d. 3C3C e. 403C
12. Suppose the memory cells at addresses 00 through 03 in the machine described in Appendix C contain the following bit patterns:

Address	Contents
00	26
01	55
02	C0
03	00

 - a. Translate the first instruction into English.
 - b. If the machine is started with its program counter containing 00, what bit pattern is in register 6 when the machine halts?

13. Suppose the memory cells at addresses 00 through 02 in the machine described in Appendix C contain the following bit patterns:

Address	Contents
00	12
01	21
02	34

- a. What would be the first instruction executed if we started the machine with its program counter containing 00?
- b. What would be the first instruction executed if we started the machine with its program counter containing 01?

14. Suppose the memory cells at addresses 00 through 05 in the machine described in Appendix C contain the following bit patterns:

Address	Contents
00	12
01	02
02	32
03	42
04	C0
05	00

When answering the following questions, assume that the machine starts with its program counter equal to 00.

- a. Translate the instructions that are executed into English.
- b. What bit pattern is in the memory cell at address 42 when the machine halts?
- c. What bit pattern is in the program counter when the machine halts?

15. Suppose the memory cells at addresses 00 through 09 in the machine described in Appendix C contain the following bit patterns:

Address	Contents
00	1C
01	03
02	2B
03	03
04	5A
05	BC
06	3A
07	00
08	C0
09	00

- Assume that the machine starts with its program counter containing 00.
- a. What will be in the memory cell at address 00 when the machine halts?
 - b. What bit pattern will be in the program counter when the machine halts?

16. Suppose the memory cells at addresses 00 through 07 in the machine described in Appendix C contain the following bit patterns:

Address	Contents
00	2B
01	07
02	3B
03	06
04	C0
05	00
06	00
07	23

- a. List the addresses of the memory cells that contain the program that will be executed if we start the machine with its program counter containing 00.
- b. List the addresses of the memory cells that are used to hold data.

17. Suppose the memory cells at addresses 00 through 0D in the machine described in Appendix C contain the following bit patterns:

Address	Contents
00	20
01	04
02	21
03	01
04	40
05	12
06	51
07	12
08	B1
09	0C
0A	B0
0B	06
0C	C0
0D	00

- Assume that the machine starts with its program counter containing 00.
- a. What bit pattern will be in register 0 when the machine halts?

- b. What bit pattern will be in register 1 when the machine halts?
- c. What bit pattern is in the program counter when the machine halts?

18. Suppose the memory cells at addresses F0 through FD in the machine described in Appendix C contain the following (hexadecimal) bit patterns:

Address	Contents
F0	20
F1	00
F2	22
F3	02
F4	23
F5	04
F6	B3
F7	FC
F8	50
F9	02
FA	B0
FB	F6
FC	C0
FD	00

If we start the machine with its program counter containing F0, what is the value in register 0 when the machine finally executes the halt instruction at location FC?

19. If the machine in Appendix C executes an instruction every microsecond (a millionth of a second), how long does it take to complete the program in Problem 18?
20. Suppose the memory cells at addresses 20 through 28 in the machine described in Appendix C contain the following bit patterns:

Address	Contents
20	12
21	20
22	32
23	30
24	B0
25	21
26	24
27	C0
28	00

Assume that the machine starts with its program counter containing 20.

- a. What bit patterns will be in registers 0, 1, and 2 when the machine halts?
- b. What bit pattern will be in the memory cell at address 30 when the machine halts?
- c. What bit pattern will be in the memory cell at address B0 when the machine halts?

21. Suppose the memory cells at addresses AF through B1 in the machine described in Appendix C contain the following bit patterns:

Address	Contents
AF	B0
B0	B0
B1	AF

What would happen if we started the machine with its program counter containing AF?

22. Suppose the memory cells at addresses 00 through 05 in the machine described in Appendix C contain the following (hexadecimal) bit patterns:

Address	Contents
00	25
01	B0
02	35
03	04
04	C0
05	00

If we start the machine with its program counter containing 00, when does the machine halt?

23. In each of the following cases, write a short program in the machine language described in Appendix C to perform the requested activities. Assume that each of your programs is placed in memory starting at address 00.
- a. Move the value at memory location D8 to memory location B3.
 - b. Interchange the values stored at memory locations D8 and B3.
 - c. If the value stored in memory location 44 is 00, then place the value 01 in memory location 46; otherwise, put the value FF in memory location 46.

24. A game that used to be popular among computer hobbyists is core wars—a variation of battleship. (The term *core* originates from an early memory technology in which 0s and 1s were represented as magnetic fields in little rings of magnetic material. The rings were called cores.) The game is played between two opposing programs, each stored in different locations of the same computer's memory. The computer is assumed to alternate between the two programs, executing an instruction from one followed by an instruction from the other. The goal of each program is to cause the other to malfunction by writing extraneous data on top of it; however, neither program knows the location of the other.

- Write a program in the machine language of Appendix C that approaches the game in a defensive manner by being as small as possible.
- Write a program in the language of Appendix C that tries to avoid any attacks from the opposing program by moving to different locations. More precisely, beginning at location 00, write a program that will copy itself to location 70 and then jump to location 70.
- Extend the program in (b) to continue relocating to new memory locations. In particular, make your program move to location 70, then to E0 (= 70 + 70) then to 60 (= 70 + 70 + 70) etc.

25. Write a program in the machine language of Appendix C to compute the sum of floating-point values stored at memory locations A0, A1, A2, and A3. Your program should store the total at memory location A4.

26. Suppose the memory cells at addresses 00 through 05 in the machine described in Appendix C contain the following (hexadecimal) bit patterns:

Address	Contents
00	20
01	C0
02	30
03	04
04	00
05	00

What happens if we start the machine with its program counter containing 00?

27. What happens if the memory cells at addresses 08 and 09 of the machine described in Appendix C contain the bit patterns B0 and 08, respectively, and the machine is started with its program counter containing the value 08?

28. Suppose the following program, written in the machine language of Appendix C, is stored in main memory beginning at address 30 (hexadecimal). What task will the program perform when executed?

```

2003
2101
2200
2310
1400
3410
5221
5331
3239
333B
B248
B038
C000

```

29. Summarize the steps involved when the machine described in Appendix C performs an instruction with op-code B. Express your answer as a set of directions as though you were telling the CPU what to do.

***30.** Summarize the steps involved when the machine described in Appendix C performs an instruction with op-code 5. Express your answer as a set of directions as though you were telling the CPU what to do.

***31.** Summarize the steps involved when the machine described in Appendix C performs an instruction with op-code 6. Express your answer as a set of directions as though you were telling the CPU what to do.

***32.** Suppose the registers 4 and 5 in the machine described in Appendix C contain the bit patterns 3A and C8, respectively. What bit pattern is left in register 0 after executing each of the following instructions:

- 5045
- 6045
- 7045
- 8045
- 9045

- *44. Write a program in the machine language of Appendix C that reverses the contents of the memory cell at address 8C. (That is, the final bit pattern at address 8C when read from left to right should agree with the original pattern when read from right to left.)
- *45. Write a program in the machine language of Appendix C that subtracts the value stored at A1 from the value stored at address A2 and places the result at address A0. Assume that the values are encoded in two's complement notation.
- *46. High definition video can be delivered at a rate of 30 frames per second (fps) where each frame has a resolution of 1920 x 1080 pixels using 24 bits per pixel. Can an uncompressed video stream of this format be sent over a USB 1.1 serial port? USB 2.0 serial port? USB 3.0 serial port? (Note: The maximum speeds of USB 1.1, USB 2.0, and USB 3.0 serial ports are 12Mbps, 480Mbps, and 5Gbps respectively.)
- *47. Suppose a person is typing forty words per minute at a keyboard. (A word is considered to be five characters.) If a machine executes 500 instructions every microsecond (millionth of a second), how many instructions does the machine execute during the time between the typing of two consecutive characters?
- *48. How many bits per second must a keyboard transmit to keep up with a typist typing forty words per minute? (Assume each character is encoded in ASCII and each word consists of six characters.)
- *49. Suppose the machine described in Appendix C communicates with a printer using the technique of memory-mapped I/O. Suppose also that address FF is used to send characters to the printer, and address FE is used to receive information about the printer's status. In particular, suppose the least significant bit at the address FE indicates whether the printer is ready to receive another character (with a 0 indicating "not ready" and a 1 indicating "ready"). Starting at address 00, write a machine language routine that waits until the printer is ready for another character and then sends the character represented by the bit pattern in register 5 to the printer.
- *50. Write a program in the machine language described in Appendix C that places 0s in all the memory cells from address A0 through C0 but is small enough to fit in the memory cells from address 00 through 13 (hexadecimal).
- *51. Suppose a machine has 200 GB of storage space available on a hard disk and receives data over a broadband connection at the rate of 15 Mbps. At this rate, how long will it take to fill the available storage space?
- *52. Suppose a satellite system is being used to receive a serial data stream at 250 Kbps. If a burst of atmospheric interference lasts 6.96 seconds, how many data bits will be affected?
- *53. Suppose you are given 32 processors, each capable of finding the sum of two multidigit numbers in a millionth of a second. Describe how parallel processing techniques can be applied to find the sum of 64 numbers in only six-millionths of a second. How much time does a single processor require to find this same sum?
- *54. Summarize the difference between a CISC architecture and a RISC architecture.
- *55. Identify two approaches to increasing throughput.
- *56. Describe how the average of a collection of numbers can be computed more rapidly with a multiprocessor machine than a single processor machine.
- *57. Write and test a Python script that reads in a floating-point radius of a circle and outputs the circumference and area of the circle.
- *58. Write and test a Python script that reads in a character string and an integer and outputs the character string repeated the number of times given by the integer.
- *59. Write and test a Python script that reads in two floating-point side lengths of a right triangle and outputs the hypotenuse length, perimeter, and area.

- *33.** Using the machine language described in Appendix C, write programs to perform each of the following tasks:
- Copy the bit pattern stored in memory location 44 into memory location AA.
 - Change the least significant 4 bits in the memory cell at location 34 to 0s while leaving the other bits unchanged.
 - Copy the least significant 4 bits from memory location A5 into the least significant 4 bits of location A6 while leaving the other bits at location A6 unchanged.
 - Copy the least significant 4 bits from memory location A5 into the most significant 4 bits of A5. (Thus, the first 4 bits in A5 will be the same as the last 4 bits.)
- *34.** Perform the indicated operations:
- | | |
|------------------------------------|------------------------------------|
| a. 111001
<u>AND 101001</u> | b. 000101
<u>AND 101010</u> |
| c. 001110
<u>AND 010101</u> | d. 111011
<u>AND 110111</u> |
| e. 111001
<u>OR 101001</u> | f. 010100
<u>OR 101010</u> |
| g. 000100
<u>OR 010101</u> | h. 101010
<u>OR 110101</u> |
| i. 111001
<u>XOR 101001</u> | j. 000111
<u>XOR 101010</u> |
| k. 010000
<u>XOR 010101</u> | l. 111111
<u>XOR 110101</u> |
- *35.** Identify both the mask and the logical operation needed to accomplish each of the following objectives:
- Put 1s in the upper 4 bits of an 8-bit pattern without disturbing the other bits.
 - Complement the most significant bit of an 8-bit pattern without changing the other bits.
 - Complement a pattern of 8 bits.
 - Put a 0 in the least significant bit of an 8-bit pattern without disturbing the other bits.
 - Put 1s in all but the most significant bit of an 8-bit pattern without disturbing the most significant bit.
 - Filter out all of the green color component from an RGB bitmap image pixel in which the middle 8 bits of a 24-bit pattern store the green information.
 - Invert all of the bits in a 24-bit RGB bitmap pixel.
 - Set all the bits in a 24-bit RGB bitmap pixel to 1, indicating the color "white".
- *36.** Write and test short Python scripts to implement each of the parts of the previous question.
- *37.** Identify a logical operation (along with a corresponding mask) that, when applied to an input string of 8 bits, produces an output string of all 0s if and only if the input string is 10000001.
- *38.** Write and test a short Python script to implement the previous question.
- *39.** Describe a sequence of logical operations (along with their corresponding masks) that, when applied to an input string of 8 bits, produces an output byte of all 0s if the input string both begins and ends with 1s. Otherwise, the output should contain at least one 1.
- *40.** Write and test a short Python script to implement the previous question.
- *41.** What would be the result of performing a 4-bit left circular shift on the following bit patterns?
- | | | |
|-----------|-------------|--------|
| a. 10101 | b. 11110000 | c. 001 |
| d. 101000 | e. 00001 | |
- *42.** What would be the result of performing a 2-bit right circular shift on the following bytes represented in hexadecimal notation (give your answers in hexadecimal notation)?
- | | |
|-------|-------|
| a. 3F | b. 0D |
| c. FF | d. 77 |
- *43.**
- What single instruction in the machine language of Appendix C could be used to accomplish a 5-bit right circular shift of register B?
 - What single instruction in the machine language of Appendix C could be used to accomplish a 2-bit left circular shift of register B?

- *44. Write a program in the machine language of Appendix C that reverses the contents of the memory cell at address 8C. (That is, the final bit pattern at address 8C when read from left to right should agree with the original pattern when read from right to left.)
- *45. Write a program in the machine language of Appendix C that subtracts the value stored at A1 from the value stored at address A2 and places the result at address A0. Assume that the values are encoded in two's complement notation.
- *46. High definition video can be delivered at a rate of 30 frames per second (fps) where each frame has a resolution of 1920 x 1080 pixels using 24 bits per pixel. Can an uncompressed video stream of this format be sent over a USB 1.1 serial port? USB 2.0 serial port? USB 3.0 serial port? (Note: The maximum speeds of USB 1.1, USB 2.0, and USB 3.0 serial ports are 12Mbps, 480Mbps, and 5Gbps respectively.)
- *47. Suppose a person is typing forty words per minute at a keyboard. (A word is considered to be five characters.) If a machine executes 500 instructions every microsecond (millionth of a second), how many instructions does the machine execute during the time between the typing of two consecutive characters?
- *48. How many bits per second must a keyboard transmit to keep up with a typist typing forty words per minute? (Assume each character is encoded in ASCII and each word consists of six characters.)
- *49. Suppose the machine described in Appendix C communicates with a printer using the technique of memory-mapped I/O. Suppose also that address FF is used to send characters to the printer, and address FE is used to receive information about the printer's status. In particular, suppose the least significant bit at the address FE indicates whether the printer is ready to receive another character (with a 0 indicating "not ready" and a 1 indicating "ready"). Starting at address 00, write a machine language routine that waits until the printer is ready for another character and then sends the character represented by the bit pattern in register 5 to the printer.
- *50. Write a program in the machine language described in Appendix C that places 0s in all the memory cells from address A0 through C0 but is small enough to fit in the memory cells from address 00 through 13 (hexadecimal).
- *51. Suppose a machine has 200 GB of storage space available on a hard disk and receives data over a broadband connection at the rate of 15 Mbps. At this rate, how long will it take to fill the available storage space?
- *52. Suppose a satellite system is being used to receive a serial data stream at 250 Kbps. If a burst of atmospheric interference lasts 6.96 seconds, how many data bits will be affected?
- *53. Suppose you are given 32 processors, each capable of finding the sum of two multidigit numbers in a millionth of a second. Describe how parallel processing techniques can be applied to find the sum of 64 numbers in only six-millionths of a second. How much time does a single processor require to find this same sum?
- *54. Summarize the difference between a CISC architecture and a RISC architecture.
- *55. Identify two approaches to increasing throughput.
- *56. Describe how the average of a collection of numbers can be computed more rapidly with a multiprocessor machine than a single processor machine.
- *57. Write and test a Python script that reads in a floating-point radius of a circle and outputs the circumference and area of the circle.
- *58. Write and test a Python script that reads in a character string and an integer and outputs the character string repeated the number of times given by the integer.
- *59. Write and test a Python script that reads in two floating-point side lengths of a right triangle and outputs the hypotenuse length, perimeter, and area.

Chapter Review Problems

(Asterisked problems are associated with optional sections.)

1. List four activities of a typical operating system.
2. What led to the development of the interactive operating system?
3. Suppose three items R, S, and T are placed in a queue in that order. Then one item is removed from the queue before a fourth item, X, is placed in the queue. Then one item is removed from the queue, the items Y and Z are placed in the queue, and then the queue is emptied by removing one item at a time. List all the items in the order in which they were removed.
4. What is the significance of the window manager in current GUI shells?
5. What is a real-time operating system?
6. If you have a PC, identify some situations in which you can take advantage of its multitasking capabilities.
7. On the basis of a computer system with which you are familiar, identify two units of application software and two units of utility software. Then explain why you classified them as you did.
8.
 - a. What is the role of the user interface of an operating system?
 - b. What is the role of the kernel of an operating system?
9. What is the use of the process table in program execution?
10. Give some examples of functions that are provided by the Python module "os".
11. What is the Linux or UNIX command for creating new processes?
12. What is the difference between a process that is ready and a process that is waiting?
13. What is the difference between virtual memory and main memory?
14. Suppose a computer contained 512MB (MiB) of main memory, and an operating system needed to create a virtual memory of twice that size using pages of 2KB (KiB). How many pages would be required?
15. What complications could arise in a time-sharing/multitasking system if two processes require access to the same file at the same time? Are there cases in which the file manager should grant such requests? Are there cases in which the file manager should deny such requests?
16. How is firmware different from hardware and software? What do you mean by a firmware update?
17. Define load balancing and scaling in the context of multiprocessor architectures.
18. What is a context switch?
19. What are the flaws of privilege level control?
20. If you have a PC, record the sequence activities that you can observe when you turn it on. Then determine what messages appear on the computer screen before the booting process actually begins. What software writes these messages?
21. Suppose a multiprogramming operating system allocated time slices of 10 milliseconds and the machine executed an average of five instructions per nanosecond. How many instructions could be executed in a single time slice?
22. If a typist types 60 words per minute (where a word is considered five characters), how much time would pass between typing each character? If a multiprogramming operating system allocated time slices in 10 millisecond units and we ignore the time required for process switches, how many time slices could be allocated between characters being typed?
23. Suppose a multiprogramming operating system is allotting time slices of 50 milliseconds. If it normally takes 8 milliseconds to position a disk's read/write head over the desired track and another 17 milliseconds for the desired data to rotate around to the read/write head, how much of a program's time slice can be spent waiting for a read operation from a disk to take place? If the machine is capable of executing 10 instructions each nanosecond,

how many instructions can be executed during this waiting period? (This is why when a process performs an operation with a peripheral device, a multiprogramming system terminates that process's time slice and allows another process to run while the first process is waiting for the services of the peripheral device.)

24. What is the drawback of using the set and clear flag system while allocating devices?
25. A process is said to be I/O-bound if it requires a lot of I/O operations, whereas a process that consists of mostly computations within the CPU/memory system is said to be compute-bound. If both a compute-bound process and an I/O-bound process are waiting for a time slice, which should be given priority? Why?
26. Would greater throughput be achieved by a system running two processes in a multiprogramming environment if both processes were I/O-bound (refer to problem 25) or if one were I/O-bound and the other were compute-bound? Why?
27. Write a set of directions that tells an operating system's dispatcher what to do when a process's time slice is over.
28. What are the various functions of the memory manager in an operating system?
29. Identify a situation in a multiprogramming system in which a process does not consume the entire time slice allocated to it.
30. What is meant by an interrupt handler in multiprogramming systems and what is its significance?
31. Answer each of the following in terms of an operating system that you use:
 - a. How do you ask the operating system to copy a file from one location to another?
 - b. How do you ask the operating system to show you the directory on a disk?
 - c. How do you ask the operating system to execute a program?
32. Answer each of the following in terms of an operating system that you use:
 - a. How does the operating system restrict access to only those who are approved users?
 - b. How do you ask the operating system to show you what processes are currently in the process table?

c. How do you tell the operating system that you do not want other users of the machine to have access to your files?

- *33. Explain an important use for the test-and-set instruction found in many machine languages. Why is it important for the entire test-and-set process to be implemented as a single instruction?
- *34. A banker with only \$100,000 loans \$50,000 to each of two customers. Later, both customers return with the story that before they can repay their loans they must each borrow another \$10,000 to complete the business deals in which their previous loans are involved. The banker resolves this deadlock by borrowing the additional funds from another source and passing on this loan (with an increase in the interest rate) to the two customers. Which of the three conditions for deadlock has the banker removed?
- *35. Students who want to enroll in Model Railroad-ing II at the local university are required to obtain permission from the instructor and pay a laboratory fee. The two requirements are fulfilled independently in either order and at different locations on campus. Enrollment is limited to 20 students; this limit is maintained by both the instructor, who will grant permission to only 20 students, and the financial office, which will allow only 20 students to pay the laboratory fee. Suppose that this registration system has resulted in 19 students having successfully registered for the course, but with the final space being claimed by two students—one who has only obtained permission from the instructor and another who has only paid the fee. Which requirement for deadlock is removed by each of the following solutions to the problem?
 - a. Both students are allowed in the course.
 - b. The class size is reduced to 19, so neither of the two students is allowed to register for the course.
 - c. The competing students are both denied entry to the class and a third student is given the twentieth space.
 - d. It is decided that the only requirement for entry into the course is the payment of the fee. Thus the student who has paid the fee gets into the course, and entry is denied to the other student.

- *36. Since each area on a computer's display can be used by only one process at a time (otherwise the image on the screen would be unreadable), these areas are nonsharable resources that are allocated by the window manager. Which of the three conditions necessary for deadlock does the window manager remove in order to avoid deadlock?
- *37. Suppose each nonsharable resource in a computer system is classified as a level 1, level 2, or level 3 resource. Moreover, suppose each process in the system is required to request the resources it needs according to this classification. That is, it must request all the required level 1 resources at once before requesting any level 2 resources. Once it receives the level 1 resources, it can request all the required level 2 resources, and so on. Can deadlock occur in such a system? Why or why not?
- *38. Each of two robot arms is programmed to lift assemblies from a conveyor belt, test them for tolerances, and place them in one of two bins depending on the results of the test. The assemblies arrive one at a time with a sufficient interval between them. To keep both arms from trying to grab the same assembly, the computers controlling the arms share a common memory cell. If an arm is available as an assembly approaches, its controlling computer reads the value of the common cell. If the value is nonzero, the arm lets the assembly pass. Otherwise, the controlling computer places a nonzero value in the memory cell, directs the arm to pick up the assembly, and places the value 0 back into the memory cell after the action is complete. What sequence of events could lead to a tug-of-war between the two arms?
- *39. Why is disabling the interrupts in a multicore operating system not considered to be an efficient approach?
- *40. A process that is waiting for a time slice is said to suffer **starvation** if it is never given a time slice.
- The pavement in the middle of an intersection can be considered as a nonsharable resource for which cars approaching the intersection compete. A traffic light rather than an operating system is used to control the allocation of the resource. If the light is able to sense the amount of traffic arriving from each direction and is programmed to give the green light to the heavier traffic, the lighter traffic might suffer from starvation. How is starvation avoided?
 - In what sense can a process starve if the dispatcher always assigns time slices according to a priority system in which the priority of each process remains fixed? (Hint: What is the priority of the process that just completed its time slice in comparison to the processes that are waiting, and consequently which routine gets the next time slice?) How, would you guess, do many operating systems avoid this problem?
- *41. Why can't special-purpose registers restrict a process in its allotted memory area? Explain your answer.
- *42. The following is the "dining philosophers" problem that was originally proposed by E. W. Dijkstra and is now a part of computer science folklore. Five philosophers are sitting at a round table. In front of each is a plate of spaghetti. There are five forks on the table, one between each plate. Each philosopher wants to alternate between thinking and eating. To eat, a philosopher requires possession of both the forks that are adjacent to the philosopher's plate. Identify the possibilities of deadlock and starvation (see problem 40) that are present in the dining philosophers' problem.
- *43. What problem arises as the lengths of the time slices in a multiprogramming system are made shorter and shorter? What about as they become longer and longer?
- *44. When is it preferable to use the deadlock prevention scheme, the deadlock avoidance scheme, and the deadlock detection and recovery scheme? Can you provide scenarios where more than one of these schemes can be used?
45. Identity two activities that can be performed by an operating system's administrator but not by a typical user.
46. How is the read action different from the write action when multiple processes access the same file?
47. Suppose a password consisted of a string of nine characters from the English alphabet

(26 characters). If each possible password could be tested in a millisecond, how long would it take to test all possible passwords?

48. Why are CPUs that are designed for multitasking operating systems capable of operating at different privilege levels?
49. How are privileged instructions handled in nonprivileged mode?
50. Identify three ways in which a process could challenge the security of a computer system if not prevented from doing so by the operating system.

Chapter Review Problems

(Asterisked problems are associated with optional sections.)

1. What is a protocol? Identify three protocols introduced in this chapter and describe the purpose of each.
2. What is interprocess communication?
3. What is meant by an intranet?
4. Describe the three types of ISP tiers.
5. What is the difference between a bus network and a star network?
6. What happens when two nodes using the CSMA/CD, encounter collision in the network?
7. Can collisions be completely eliminated by using the CSMA/CA like collision avoidance protocols?
8. What is the last mile problem? Describe a technique for solving it.
9. How does a hub differ from a switch?
10. How does a router differ from such devices as repeaters, bridges, and switches?
11. State the distinction between the client-server model and the P2P model.
12. What is the difference between congestion control and flow control?
13. Using 32-bit Internet addresses was originally thought to provide ample room for expansion, but that conjecture is not proving to be accurate. IPv6 uses 128-bit addressing. Will that prove to be adequate? Justify your answer. (For example, you might compare the number of possible addresses to the population of the world.)
14. Encode each of the following bit patterns using dotted decimal notation.
 - a. 000001010001001000100011
 - b. 1000000000100000
 - c. 0011000000011000
15. What bit pattern is represented by each of the following dotted decimal patterns?
 - a. 0.0
 - b. 26.19.1
 - c. 8.12.20.13
16. Suppose the address of an end system on the Internet is quoted as 134.48.4.122. What is the 32-bit address in hexadecimal notation?
17. What is on-demand streaming?
18. If a computer's mnemonic Internet address is `batman.batcave.metropolis.gov` what might you conjecture about the structure of the domain containing the machine?
19. Explain the components of the email address `kermit@animals.com`
20. In the context of VoIP, what is the difference between an analog telephone adapter and an embedded phone?
21. What is the role of a name server?
22. What is the distinction between routing and forwarding?
23. Define each of the following:
 - a. Cloud computing
 - b. Hot spot
 - c. Top-level domains
 - d. Secure Shell
51. What are the conditions that lead to a deadlock?
52. What are the policies that a user should follow to manage login passwords?
53. Give appropriate examples of auditing software and sniffing software.
54. How is security relevant in current operating systems?
55. How is the booting process different in embedded systems from traditional systems?

24. Define each of the following:
 - a. Content delivery networks
 - b. Anycast
 - c. Markup languages
25. Many “lay users” of the Internet interchange the terms *Internet* and *World Wide Web*. To what do each of the terms correctly refer?
26. When viewing a simple webpage, ask your browser to display the source version of the document. Then identify the basic structure of the document. In particular, identify the head and the body of the document and list some of the statements you find in each.
27. State the difference between sniffing and phishing.
28. Modify the HTML document below so that the word “Rover” is linked to the document whose URL is <http://animals.org/pets/dogs.html>.

```
<html>
<head>
<title>Example</title>
</head>
<body>
<h1> My Pet Dog</h1>
<p>My dog's name is Rover.</p>
</body>
</html>
```
29. Draw a sketch showing how the following HTML document would appear when displayed on a computer screen.

```
<html>
<head>
<title>Example</title>
</head>
<body>
<h1> My Pet Dog</h1>
<img src = "Rover.jpg">
</body>
</html>
```
30. Using the informal XML style presented in the text, design a markup language for representing simple algebraic expressions as text files.
31. Using the informal XML style presented in the text, design a set of tags that a word processor might use for marking the underlying text. For example, how would a word processor indicate what text should be bold, italic, underlined, and so on?
32. Using the informal XML style presented in the text, design a set of tags that could be used to mark motion picture reviews according to the way the text items should appear on a printed page. Then design a set of tags that could be used to mark the reviews according to the meaning of the items in the text.
33. Using the informal XML style presented in the text, design a set of tags that could be used to mark articles about sporting events according to the way the text items should appear on a printed page. Then design a set of tags that could be used to mark the articles according to the meaning of the items in the text.
34. Identify the components of the following URL and describe the meaning of each.
<http://lifeforms.com/animals/moviestars/kermit.html>
35. Identify the components of each of the following abbreviated URLs.
 - a. <http://www.farmtools.org/windmills.html>
 - b. <http://castles.org/>
 - c. www.coolstuff.com
36. How would the action of a browser differ if you asked it to “find the document” at the URL <http://stargazer.universe.org> as opposed to <https://stargazer.universe.org>?
37. Give two examples of client-side activities on the Web. Give two examples of server-side activities on the Web.
- *38. What is the TCP/IP protocol suite?
- *39. In a network based on the bus topology, the bus is a nonsharable resource for which the machines must compete in order to transmit messages. How is deadlock (see the optional Section 3.4) controlled in this context?
- *40. List the four layers in the Internet software hierarchy and identify a task performed by each layer.
- *41. Why does the transport layer chop large messages into small packets?

- *42. When an application asks the transport layer to use TCP to transmit a message, what additional messages will be sent by the transport layer in order to fulfill the application layer's request?
- *43. In what way could TCP be considered a better protocol for implementing the transport layer than UDP? In what way could UDP be considered better than TCP?
- *44. What does it mean to say that the TCP is a connection-oriented protocol?
- *45. At what layer in the TCP/IP protocol hierarchy could a firewall be placed to filter incoming traffic by means of
 - a. Message content
 - b. Source address
 - c. Type of application
- 46. Suppose you wanted to establish a firewall to filter out email messages containing certain terms and phrases. Would this firewall be placed at your domain's gateway or at the domain's mail server? Explain your answer.
- 47. What is SSL and what are its benefits?
- 48. What is the significance of using spam filters?
- 49. What is the significance of certificate authorities in public-key encryption?
- 50. In what sense does the global nature of the Internet limit legal solutions to Internet problems?

Chapter Review Problems

(Asterisked problems are associated with optional sections.)

1. Give an example of a set of steps that conforms to the informal definition of an algorithm given in the opening paragraph of Section 5.1 but does not conform to the formal definition given in Figure 5.1.
2. Explain the distinction between an ambiguity in a proposed algorithm and an ambiguity in the representation of an algorithm.
3. Describe how the use of primitives helps remove ambiguities in an algorithm's representation.
4. Select a subject with which you are familiar and design a pseudocode for giving directions in that subject. In particular, describe the primitives you would use and the syntax you would use to represent them. (If you are having trouble thinking of a subject, try sports, arts, or crafts.)
5. Does the following program represent an algorithm in the strict sense? Why or why not?


```
Count = 0
while (Count != 5):
    Count = Count + 2
```
6. In what sense do the following three steps not constitute an algorithm?

Step 1: Draw a circle with center coordinates (2, 5) and radius 3.

Step 2: Draw a circle with center coordinates (6, 5) and radius 5.

Step 3: Draw a line segment whose endpoints are at the intersections of the previous two circles.

7. Rewrite the following program segment using a **repeat** structure rather than a **while** structure. Be sure the new version prints the same values as the original. Initialization:

```
num = 0
while (num < 50):
    if (num is Odd)
        print(num is Odd)
    num = num + 1
```

8. Rewrite the following program segment using a **while** structure rather than a **repeat** structure. Be sure the new version prints the same values as the original.

```
num = 100
repeat:
    print(num)
    num = num - 1
until (num > 0)
```

9. What must be done to translate a posttest loop expressed in the form

```
repeat:
    (. . . )
until (. . . )
```

into an equivalent posttest loop expressed in the form

```
do:
    (. . . )
while (. . . )
```

10. Design an algorithm that, when given an arrangement of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, rearranges the digits so that the new arrangement represents the next larger value that can be represented by these digits (or reports that no such rearrangement exists if no rearrangement produces a larger value). Thus 5647382901 would produce 5647382910.
11. Design an algorithm for finding all the factors of a positive integer. For example, in the case of the integer 12, your algorithm should report the values 1, 2, 3, 4, 6, and 12.
12. Design an algorithm for determining whether a particular year is a leap year. For example, the year 2000 was a leap year.

13. What is the difference between a formal programming language and a pseudocode?
14. What is the difference between syntax and semantics?
15. The following is an addition problem in traditional base 10 notation. Each letter represents a different digit. What digit does each letter represent? How did you get your foot in the door?

$$\begin{array}{r} \text{SEND} \\ + \text{MORE} \\ \hline \text{MONEY} \end{array}$$

16. The following is a multiplication problem in traditional base 10 notation. Each letter represents a different digit. What digit does each letter represent? How did you get your foot in the door?

$$\begin{array}{r} \text{XY} \\ \times \text{YX} \\ \hline \text{XY} \\ \text{YZ} \\ \hline \text{WVY} \end{array}$$

17. The following is a multiplication problem in binary notation. Each letter represents a unique binary digit. Which letter represents 1 and which represents 0? Design an algorithm for solving problems like this.

$$\begin{array}{r} \text{XXYX} \\ \times \text{X XY} \\ \hline \text{XXYXY} \end{array}$$

18. Four prospectors with only one lantern must walk through a mine shaft. At most, two prospectors can travel together and any prospector in the shaft must be with the lantern. The prospectors, named Andrews, Blake, Johnson, and Kelly, can walk through the shaft in one minute, two minutes, four minutes, and eight minutes, respectively. When two walk together they travel at the speed of the slower prospector. How can all four prospectors get through the mine shaft in only 15 minutes? After you have solved this problem, explain how you got your foot in the door.
19. Starting with a large wine glass and a small wine glass, fill the small glass with wine and then pour that wine into the large glass. Next, fill the small glass with water and pour

some of that water into the large glass. Mix the contents of the large glass, and then pour the mixture back into the small glass until the small glass is full. Will there be more water in the large glass than there is wine in the small glass? After you have solved this problem, explain how you got your foot in the door.

20. Two bees, named Romeo and Juliet, live in different hives but have met and fallen in love. On a windless spring morning, they simultaneously leave their respective hives to visit each other. Their routes meet at a point 50 meters from the closest hive, but they fail to see each other and continue on to their destinations. At their destinations, they spend the same amount of time to discover that the other is not home and begin their return trips. On their return trips, they meet at a point that is 20 meters from the closest hive. This time they see each other and have a picnic lunch before returning home. How far apart are the two hives? After you have solved this problem, explain how you got your foot in the door.

21. Design an algorithm that, given two strings of characters, tests whether the second string is same as the first string or not.

22. The following algorithm is designed to print the beginning of what is known as the Fibonacci sequence. Identify the body of the loop. Where is the initialization step for the loop control? The modification step? The test step? What list of numbers is produced?

```
Last = 0
Current = 1
while (Current < 100):
    print(Current)
    Temp = Last
    Last = Current
    Current = Last + Temp
```

23. What sequence of numbers is printed by the following algorithm if the input value is 1?

```
def CodeWrite (num):
    While (num < 100):
        for (i = 2; i <= num/2; ++i)
            if (num % i == 0) flag = 1;
            if (flag == 0) print (num);
            num ++;
```

24. Modify the function `CodeWrite` in the preceding problem so that perfect square values are printed.

25. What letters are interrogated by the binary search (Figure 5.14) if it is applied to the list A, B, C, D, E, F, G, H, I, J, K, L, M, N, O when searching for the value J? What about searching for the value Z?

26. After performing many sequential searches on a list of 6,000 entries, what would you expect to be the average number of times that the target value would have been compared to a list entry? What if the search algorithm was the binary search?

27. Identify the termination condition in each of the following iterative statements.

a. `while (Count < 5):`

. . .

b. `repeat:`

. . .

`until (Count == 1)`

c. `while ((Count < 5) and (Total < 56)):`

. . .

28. Identify the body of the following loop structure and count the number of times it will be executed. What happens if the test is changed to read “(Count != 6)”?

```
Count = 1
while (Count != 7):
    print(Count)
    Count = Count + 3
```

29. What problems do you expect to arise if the following program is implemented on a computer? (Hint: Remember the problem of round-off errors associated with floating-point arithmetic.)

```
Count = one_tenth
repeat:
    print(Count)
    Count = Count + one_tenth
until (Count == 1)
```

30. Design a recursive version of the Euclidean algorithm (question 3 of Section 5.2).

31. Suppose we apply both `Check1` and `Check2` (defined next) to the input value 1. What is the difference in the printed output of the two routines?

```
def Check1 (num):
    if (num % 2 == 0):
        print(num)
        Check1(num + 1)
def Check2(num):
    if (num % 2 == 1):
        print(num)
        Check2(num + 1)
```

32. Identify the important constituents of the control mechanism in the routines of the previous problem. In particular, what condition causes the process to terminate? Where is the state of the process modified toward this termination condition? Where is the state of the control process initialized?

33. Identify the termination condition in the following recursive function.

```
def XXX (N):
    if (N == 5):
        XXX(N + 1)
```

34. Call the function `CodeWrite` (defined below) with the value 100 and record the values that are printed.

```
def CodeWrite (N):
    if (N > 0):
        print(N)
        CodeWrite(N / 2)
    print(N + 1)
```

35. Call the function `MysteryPrint` (defined below) with the value 2 and record the values that are printed.

```
def MysteryPrint (N):
    if (N > 0):
        print(N)
        MysteryPrint(N - 2)
    else:
        print(N)
        if (N > -1):
            MysteryPrint(N + 1)
```

36. Design an algorithm to generate the sequence of positive integers (in increasing order) whose only prime divisors are 2 and 3; that is, your program should produce the sequence 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 27, Does your program represent an algorithm in the strict sense?

37. Answer the following questions in terms of the list: 19, 37, 53, 71, 96, 137, 289, 374, 559, 797, 979.

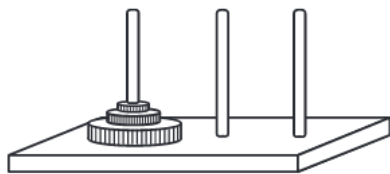
- Which search algorithm (sequential or binary) will find the number 137 more quickly?
- Which search algorithm (sequential or binary) will find the number 19 more quickly?
- Which search algorithm (sequential or binary) will detect the absence of the number 99 more quickly?
- Which search algorithm (sequential or binary) will detect the absence of the number 111 more quickly?
- How many numbers will be interrogated when searching for the number 96 when using the sequential search? How many will be interrogated when using the binary search?

38. A positive integer is called an Armstrong number if the sum of the cubes of individual digits of the number is equal to that number itself. For example, the sum of cubes of individual digits of the number 153 is $1^3 + 5^3 + 3^3 = 153$. Hence, the number 153 is called an Armstrong number. Design an algorithm that checks whether a given number is an Armstrong number or not.

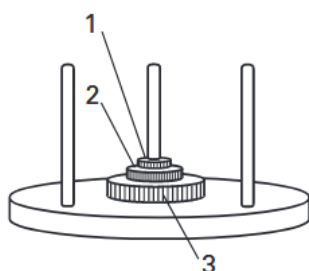
39. a. Suppose you must sort a list of ten integers in descending order, and you have already designed an algorithm that does this for six integers. Design an algorithm to sort ten integers by taking advantage of the previously designed algorithm.
- b. Design a recursive algorithm to sort arbitrary lists of integers in descending order based on the technique used in (a).

40. The puzzle called the Towers of Hanoi consists of three pegs, one of which contains several rings stacked in order of descending diameter from bottom to top. The problem is to move the stack of rings to another peg. You are allowed to move only one ring at a time, and at no time is a ring to be placed on top of a smaller one. Observe that if the puzzle involved only one ring, it would be extremely easy. Moreover, when faced with the problem of moving several rings, if you could move all but the largest ring to another peg, the largest ring could then be placed on the third peg, and then the problem would be to move the remaining rings on top of it. Using this observation, develop a recursive algorithm

for solving the Towers of Hanoi puzzle for an arbitrary number of rings.



41. Another approach to solving the Towers of Hanoi puzzle (question 40) is to imagine the pegs arranged on a circular stand with a peg mounted at each of the positions of 4, 8, and 12 o'clock. The rings, which begin on one of the pegs, are numbered 1, 2, 3, and so on, starting with the smallest ring being 1. Odd-numbered rings, when on top of a stack, are allowed to move clockwise to the next peg; likewise, even-numbered rings are allowed to move counterclockwise (as long as that move does not place a ring on a smaller one). Under this restriction, always move the largest-numbered ring that can be moved. Based on this observation, develop a nonrecursive algorithm for solving the Towers of Hanoi puzzle.



42. Develop two algorithms, one based on a loop structure and the other on a recursive structure, to print the daily salary of a worker who each day is paid twice the previous day's salary (starting with one penny for the first day's work) for a 30-day period. What problems relating to number storage are you likely to encounter if you implement your solutions on an actual machine?
43. Design an algorithm to find the square root of a positive number by starting with the number itself as the first guess and repeatedly producing a new guess from the previous one by averaging the previous guess with the result of dividing the original number by the previous guess. Analyze the control of this repetitive process. In particular, what condition should terminate the repetition?

44. Design an algorithm that lists all possible 4-digit integers made from digits 1 to 5 and are less than 5000.
45. Design an algorithm that, given a list of names, finds the longest name in the list. Use the **for** loop structure. Determine what your solution does if there are several "longest" names in the list. In particular, what would your algorithm do if all the names had the same length?
46. Design an algorithm that, given a list of nine or more names, sorts the list in alphabetical order and finds the name with the minimum number of characters.
47. Arrange the names Brenda, Doris, Raymond, Steve, Timothy, and William in an order that requires the least number of comparisons when sorted by the insertion sort algorithm (Figure 5.11).
48. What is the largest number of entries that are interrogated if the binary search algorithm (Figure 5.14) is applied to a list of 4,000 names? How does this compare to the sequential search (Figure 5.6)?
49. Use big-theta notation to classify the traditional grade school algorithms for addition and multiplication. That is, if asked to add two numbers each having n digits, how many individual additions must be performed? If requested to multiply two n -digit numbers, how many individual multiplications are required?
50. Sometimes a slight change in a problem can significantly alter the form of its solution. For example, find a simple algorithm for solving the following problem and classify it using big-theta notation:

Divide a group of people into two disjoint subgroups (of arbitrary size) such that the difference in the total ages of the members of the two subgroups is as large as possible.

Now change the problem so that the desired difference is as small as possible and classify your approach to the problem.

51. From a given a list of 1000 integers from 1 to 1000, extract pairs of numbers whose product is 2424. How efficient is your approach to the problem?

52. Explain what will be the values of `a` and `b` if we call the function `CodeWrite` (defined below) with inputs 78 and 89.

```
def CodeWrite (a, b):  
    if (a > 0 and b > 0):  
        a = a + b;  
        a = a + b;  
        a = a + b;  
        print(a and b)
```

53. The following program segment is designed to compute the product of two nonnegative integers `X` and `Y` by accumulating the sum of `X` copies of `Y`; that is, 3 times 4 is computed by accumulating the sum of three 4s. Is the program segment correct? Explain your answer.

```
Product = 0  
Count = 0  
repeat:  
    Product = Product + Y  
    Count = Count + 1  
until (Count == X)
```

54. The following program segment is designed to report which of the positive integers `X` and `Y` is a divisor of the other. Is the program segment correct? Explain your answer.

```
if (X < Y)  
    if (Y % X == 0):  
        print('X is Divisor of Y')  
else if (X % Y == 0):  
    print('Y is Divisor of X')
```

55. The following program segment is designed to find the smallest entry in a nonempty list of integers. Is it correct? Explain your answer.

```
TestValue = first list entry  
CurrentEntry = first list entry  
while (CurrentEntry is not the  
    last entry):  
    if (CurrentEntry < TestValue):  
        TestValue = CurrentEntry  
CurrentEntry = the next list entry
```

56. a. Identify the preconditions for the sequential search as represented in Figure 5.6. Establish a loop invariant for the `while` structure in that program that, when combined with the termination condition, implies that upon termination of the loop, the algorithm will report success or failure correctly.
b. Give an argument showing that the `while` loop in Figure 5.6 does in fact terminate.

57. The following program segment is designed to calculate the factorial of a nonnegative number `N` using recursion. Is the program segment correct? Explain your answer. Modify this program segment to calculate the factorial without using recursion. Be sure that the output values do not change.

```
int fact (int N) {  
    if (N == 0)  
        return 1;  
    return N * fact (N - 1);  
}
```

Chapter Review Problems

Asterisked problems are associated with optional sections.

1. What does it mean to say that a programming language is machine independent?
2. Translate the following Python program into the machine language described in Appendix C.

```
x = 0
while (x < 3):
    x = x + 1
```

3. Translate the statement

`Halfway = Length + Width`

into the machine language of Appendix C, assuming that `Length`, `Width`, and `Halfway` are all represented in floating-point notation.

4. Translate the high-level statement

```
if (X == 0):
    Z = Y + W
else:
    Z = Y + X
```

into the machine language of Appendix C, assuming that `W`, `X`, `Y`, and `Z` are all values represented in two's complement notation, each using one byte of memory.

5. Why was it necessary to identify the type of data associated with the variables in question 4 in order to translate the statements? Why do many high-level programming languages require the programmer to identify the type of each variable at the beginning of a program?
6. What are the different levels of programming languages?
7. Suppose the function f expects a string as its input and returns the same string as its output by removing the odd-positioned letters from the string. What is the result returned by function $f(f(\text{"programming"}))$?
8. Suppose the function f expects two numeric values as its inputs and returns their addition as its output value, and g is a function that returns the subtraction of the two values given as its input. If a and b represent numeric values, what is the result returned by $f(f(a,b), g(a,b))$?

9. Suppose you are going to write an object-oriented program for calculating grades of students. What data should be stored inside the object representing a student's grades? To what messages should that object be able to respond? What are the different objects that might be used in the program?
10. Summarize the distinction between a machine language and an assembly language.
11. Design an assembly language for the machine described in Appendix C.
12. John Programmer argues that the ability to declare constants within a program is not necessary because variables can be used instead. For example, our example of `AirportAlt` in Section 6.2 could be handled by declaring `AirportAlt` to be a variable and then assigning it the required value at the beginning of the program. Why is this not as good as using a constant?
13. Summarize the distinction between the declaration and the definition of a variable.
14. Explain the differences between a local variable and a global variable.
15. a. What is operator precedence?
b. Depending on operator precedence, what values could be associated with the expression $6 + 2 \times 3$?
16. Explain the advantages of code reuse.
17. What will be the output of the following C code? Explain your answer.

```
#define sqr(X)  X * X
main() {
    int k = sqr (10 + 20);
    printf("%d", k);
}
```

18. Draw a flowchart representing the structure expressed by the following for statement.

```
for (int x = 2; x < 8; ++x)
{ . . . }
```


19. Translate the following `while` statement into an equivalent program segment using the Python `for` statement. Initialization: `x = 1`

```
while ( x! = 100)
{ x = x + 1 }
```
20. If you are familiar with written music, analyze musical notation as a programming language. What are the control structures? What is the syntax for inserting program comments? What music notation has semantics similar to the `for` statement in Figure 6.7?
21. Rewrite the following program segment using a `while` loop instead of a `for` loop.

```
for (i = 0; i<100; i++) {
  if (i % 2 == 0)
  {
    print(i)
  }
}
```
22. Draw a flowchart representing the structure expressed by the following `if` and `else` conditional statements.

```
if (a > b) {
  if (c > a) {
    if (d > c) { print(d) }
  }
} else if (b > a) {
  if (c > b) {
    if (d > c) { print(d) }
  }
}
```
23. Summarize the following rat's-nest routine with a single if-else statement:

```
    if X > 5 then goto 80
    X = X + 1
    goto 90
80  X = X + 2
90  stop
```
24. Summarize the basic control structures found in imperative and object-oriented programming languages for performing each of the following activities:
 - a. Determining which command should be executed next
 - b. Repeating a collection of commands
 - c. Changing a variable's value
25. Summarize the distinction between code generation and code optimization.
26. Suppose the variable `P` in a program is of type integer. What will be the output of the following two program statements?

```
P = 2.5 + 3.5
print(P)
```
27. What does it mean to say that a programming language is strongly typed?
28. Can we return more than one value from a function? Explain your answer.
29. Suppose the Python function `Modify` is defined by

```
def Modify (Y):
    Y = 7
    print(Y)
```

If parameters are passed by value, what will be printed when the following program segment is executed? What if parameters are passed by reference?

```
X = 5
Modify(X)
print(X)
```
30. Suppose the Python function `Modify` is defined by

```
def Modify (Y):
    Y = 9
    print(X)
    print(Y)
```

Also suppose that `X` is a global variable. If parameters are passed by value, what will be printed when the following program segment is executed? What if parameters are passed by reference?

```
X = 5
Modify(X)
print(X)
```
31. Sometimes an actual parameter is passed to a function by producing a duplicate to be used by the function (as when the parameter is passed by value), but when the function is completed the value in the function's copy is transferred to the actual parameter before the calling function continues. In such cases

the parameter is said to be passed by value-result. What would be printed by the program segment in question 30 if parameters were passed by value-result?

32. What is the main difference between passing parameters to a function by value and passing parameters to a function by address? Explain your answer with a proper example for each.

33. What ambiguity exists in the statement

$$P = 9/4 - 1$$

34. Suppose a small company has five employees and is planning to increase the number to six. Moreover, suppose one of the company's programs contained the following assignment statements.

```
DailySalary = TotalSal/5;
AvgSalary   = TotalSal/5;
DailySales  = TotalSales/5;
AvgSales     = TotalSales/5;
```

How would the task of updating the program be simplified if the program had originally been written using constants named `NumberOfEmp` and `WorkWeek` (both set to the value 5) so that the assignment statements could be expressed as

```
DailySalary = TotalSal/DaysWk;
AvgSalary   = TotalSal/NumEmp1;
DailySales  = TotalSales/DaysWk;
AvgSales     = TotalSales/NumEmp1;
```

35. a. What is the distinction between an assembly language and a third-generation programming language?
b. Give an example of each.

36. Draw a syntax diagram representing the structure of the Python nested `if-else` statement.

37. Design a set of syntax diagrams to describe the syntax of telephone numbers in your locality. For instance, in the United States telephone numbers consist of an area code, followed by a regional code, followed by a four-digit number such as (444) 555-1234.

38. Design a set of syntax diagrams to describe simple sentences in your native language.

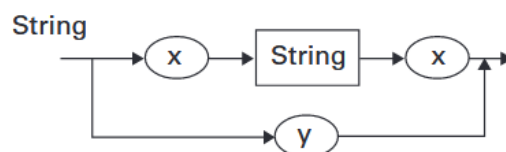
39. Design a set of syntax diagrams to describe different ways of representing full names

such as *firstname - middlename - lastname* or the reverse of it.

40. Design a set of syntax diagrams that describes the grammatical structure of "sentences" that consist of occurrences of the word *yes* followed by the same number of the word *no*. For example, "*yes yes no no*" would be such a sentence, whereas "*no yes*," "*yes no no*," and "*yes no yes*" would not.

41. Give an argument to the effect that a set of syntax diagrams cannot be designed that describes the grammatical structure of "sentences" that consist of occurrences of the word *yes*, followed by the same number of occurrences of the word *no*, followed by the same number of occurrences of the word *maybe*. For example, "*yes no maybe*" and "*yes yes no no maybe maybe*" would be such sentences, whereas "*yes maybe*," "*yes no no maybe maybe*," and "*maybe no*" would not.

42. Write a sentence describing the structure of a string as defined by the following syntax diagram. Then, draw the parse tree for the string *xyxyx*.



43. Add syntax diagrams to those in question 5 of Section 6.4 to obtain a set of diagrams that defines the structure Dance to be either a Chacha or a Waltz, where a Waltz consists of one or more copies of the pattern

forward diagonal close

or

backward diagonal close

44. Draw the parse tree for the expression $a \div d \times b \div c + a \div d \times b \div c$ based on the syntax diagrams in Figure 6.15.
45. What code optimization could be performed by a code generator when building the machine code representing the statement

```
if (X == 5):
    Z = X + 2
```

```
else:
```

```
    Z = X + 4
```

46. What will be the output of following code?

```
if (printf ("Hello")) {  
    printf("World");  
} else {  
    printf("Hello");  
}
```

47. Simplify the following program segment

```
while (Z < 10):  
    Z++;
```

48. In an object-oriented programming environment, how are types and classes similar? How are they different?

49. Describe how inheritance might be used to develop classes describing various types of sports.

50. What is the difference between the public and private parts of a class?

51. a. Give an example of a situation in which an instance variable should be private.
b. Give an example of a situation in which an instance variable should be public.
c. Give an example of a situation in which a method should be private.
d. Give an example of a situation in which a method should be public.

52. Describe some objects that might be found in a program for simulating the pedestrian traffic in a hotel lobby. Include explanations of the actions some of the objects should be able to perform.

*53. What does the term *type cast* mean in the context of a programming language?

*54. What is the difference between `while` and `do-while` loops in the context of a programming language?

*55. Draw a diagram (similar to Figure 6.25) representing the resolutions needed to show that the collection of statements $(Q \text{ OR } \neg R)$, $(T \text{ OR } R)$, $\neg P$, $(P \text{ OR } \neg T)$, and $(P \text{ OR } \neg Q)$ are inconsistent.

*56. Is the collection of statements $\neg R$, $(T \text{ OR } R)$, $(P \text{ OR } \neg Q)$, $(Q \text{ OR } \neg T)$, and $(R \text{ OR } \neg P)$ consistent? Explain your answer.

*57. Extend the Prolog program outlined in questions 3 and 4 of Section 6.7 to include the additional family relationships of uncle, aunt, grandparent, and cousin. Also add a rule that defines `parents (X, Y, Z)` to mean that X and Y are Z's parents.

*58. Assuming that the first statement in the following Prolog program is intended to mean "Alice likes sports," translate the last two statements of the program. Then, list all the things that, based on this program, Prolog would be able conclude that Alice likes. Explain your list.

```
likes(alice, sports).  
likes(alice, music).  
likes(carol, music).  
likes(david, X) :- likes(X, sports).  
likes(alice, X) :- likes(david, X).
```

*59. What problem would be encountered if the following program segment was executed on a computer in which values are represented in the eight-bit floating-point format described in Section 1.7?

```
X = 0.01  
while (X != 1.00):  
    print(X)  
    X = X + 0.01
```


Chapter Review Problems

(Asterisked problems are associated with optional sections.)

1. Draw pictures showing how the array below appears in a machine's memory when stored in row major order and in column major order:

A	B	C	D
E	F	G	H
I	J	K	L

2. Suppose an array with six rows and eight columns is stored in row major order starting at address -50 (base 10). If each entry in the array requires two memory cells, what is the address of the entry in the fifth row and seventh column? What if each entry requires three memory cells?
3. Rework question 2 assuming column major order rather than row major order.
4. Write a function such that if an element in an $M \times N$ matrix is 0, its entire row and column are set to 0.
5. Why is a contiguous list considered to be a convenient storage structure for implementing static lists, but not for implementing dynamic lists? Explain your answer.
6. Suppose you want to insert the number 3 in the list of numbers 1, 2, 4, 5, 6, 7, 8. What activities are required to insert the number 3 in the list, assuming that the order of the list is to be maintained?
7. The following table represents the contents of some cells in a computer's main memory along with the address of each cell represented. Note that some of the cells contain letters of the alphabet, and each such cell is followed by an empty cell. Place addresses in these empty cells so that each cell containing a letter together with the following cell form an entry in a linked list in which the letters appear in alphabetical order. (Use zero for the null pointer.) What address should the head pointer contain?

Address	Contents
11	C
12	
13	G
14	
15	E
16	
17	B
18	
19	U
20	
21	F
22	

8. The following table represents a portion of a linked list in a computer's main memory. Each entry in the list consists of two cells: The first contains a letter of the alphabet; the second contains a pointer to the next list entry. Alter the pointers so that the letter N is no longer in the list. Then replace the letter N with the letter G and alter the pointers so that the new letter appears in the list in its proper place in alphabetical order.

Address	Contents
30	J
31	38
32	B
33	30
34	X
35	46
36	N
37	40
38	K
39	36
40	P
41	34

9. The following table represents a linked list using the same format as in the preceding problems. If the head pointer contains the value 44, what name is represented by

the list? Change the pointers so that the list contains the name Jean.

Address	Contents
40	N
41	46
42	I
43	40
44	J
45	50
46	E
47	00
48	M
49	42
50	A
51	40

10. Which of the following routines correctly inserts **NewEntry** immediately after the entry called **PreviousEntry** in a linked list? What is wrong with the other routine?

Routine 1:

1. Copy the value in the pointer field of **PreviousEntry** into the pointer field of **NewEntry**.
2. Change the value in the pointer field of **PreviousEntry** to the address of **NewEntry**.

Routine 2:

1. Change the value in the pointer field of **PreviousEntry** to the address of **NewEntry**.
2. Copy the value in the pointer field of **PreviousEntry** into the pointer field of **NewEntry**.

11. Design a function to find the *K*th element in a single linked list with *n* elements.
12. Design a function to check whether the elements of a single linked list with *n* elements, form a palindrome.
13. Design a function for counting the nodes of a linked list.
14. a. Suppose you were given a linked list with *n* elements. Design an algorithm to remove duplicates from an unsorted linked list.

- b. If the use of a temporary buffer is not permitted to solve part (a), how would you solve the problem?

15. Sometimes a single linked list is given two different orders by attaching two pointers to each entry rather than one. Fill in the table below so that by following the first pointer after each letter one finds the name Carol, but by following the second pointer after each letter one finds the letters in alphabetical order. What values belong in the head pointer of each of the two lists represented?

Address	Contents
60	O
61	
62	
63	C
64	
65	
66	A
67	
68	
69	L
70	
71	
72	R
73	
74	

16. The table below represents a stack stored in a contiguous block of memory cells, as discussed in the text. If the base of the stack is at address 10 and the stack pointer contains the value 12, what value is retrieved by a pop instruction? What value is in the stack pointer after the pop operation?

Address	Contents
10	F
11	C
12	A
13	B
14	E

17. Draw a table showing the final contents of the memory cells if the instruction in question 16 had been to push the letter D on the stack rather than to pop a letter. What would the value in the stack pointer be after the push instruction?

18. How would you design a stack which, in addition to the traditional push and pop operations, also supports an operation called `min` which returns the element having the minimum value in the stack? `push`, `pop`, and `min` should all operate in $O(1)$ time.
19. Describe how a single array can be used to implement three stacks.
20. Suppose you were given a stack and you were allowed to use one additional stack, without copying the elements into any other data structure. Write a program to sort the stack in ascending order (biggest items on the top). The stack supports `push`, `pop`, `peek`, and `isEmpty` operations.
21. Suppose you were given three stacks and you were only allowed to move entries one at a time from one stack to another. Design an algorithm for reversing two adjacent entries on one of the stacks.
22. Suppose we want to create a stack of names that vary in length. Why is it advantageous to store the names in separate areas of memory and then build the stack out of pointers to these names rather than allowing the stack to contain the names themselves?
23. Design a function called `TwoStacks` which implements a queue by using two stacks.
24. Suppose a stack consists of boxes of specific width, height, and depth. Boxes can be stacked on the top if the underlying box is larger in all dimensions, but cannot be rotated. Try to implement the tallest stack, where the height of a stack is the sum of the heights of each box.
25. Suppose the entries in a queue require one memory cell each, the head pointer contains the value 11, and the tail pointer contains the value 17. What are the values of these pointers after one entry is inserted and two are removed?
26. The Towers of Hanoi is a classic puzzle consisting of 3 towers, and N disks of different sizes which can slide onto any tower. The puzzle starts with the disks sorted in an

ascending order from top to bottom. It has the following constraints:

- a. Only one disk can be moved at a time.
- b. A disk is moved from the top of one tower onto the next tower.
- c. A disk can only be placed on top of a larger disk.

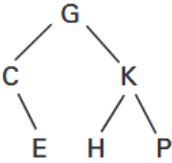
Write a program to move the disks from the first tower to the last using stacks.

27. Design a method that counts all the possible ways in which a person can run up the stairs, if he can jump one, two or three steps at a time.
28. Suppose you were given two queues and you were only allowed to move one entry at a time from the head of a queue to the tail of either. Design an algorithm for reversing two adjacent entries in one of the queues.
29. The table below represents a tree stored in a machine's memory. Each node of the tree consists of three cells. The first cell contains the data (a letter), the second contains a pointer to the node's left child, and the third contains a pointer to the node's right child. A value of 0 represents a null pointer. If the value of the root pointer is 55, draw a picture of the tree.

Address	Contents
40	G
41	0
42	0
43	X
44	0
45	0
46	J
47	49
48	0
49	M
50	0
51	0
52	F
53	43
54	40
55	W
56	46
57	52

30. The table below represents the contents of a block of cells in a computer's main memory. Note that some of the cells contain letters of the alphabet, and each of those cells is followed by two blank cells. Fill in the blank cells so that the memory block represents the tree that follows. Use the first cell following a letter as the pointer to that node's left child and the next cell as the pointer to the

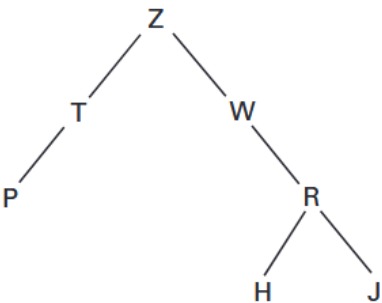
Address	Contents
30	C
31	
32	
33	H
34	
35	
36	K
37	
38	
39	E
40	
41	
42	G
43	
44	
45	P
46	
47	



right child. Use 0 for null pointers. What value should be in the root pointer?

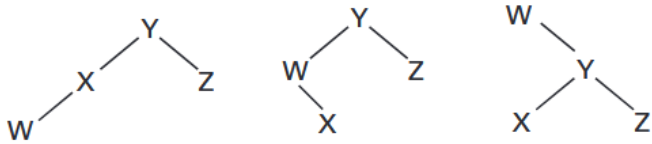
- 31. Write an algorithm to create a binary search tree with minimal height, for a given sorted array of integers.
- 32. Design a nonrecursive algorithm to replace the recursive one represented in Figure 8.24. Use a stack to control any backtracking that might be necessary.

- 33. Apply the recursive tree-printing algorithm of Figure 8.24. Draw a diagram representing the nested activations of the algorithm (and the current position in each) at the time node X is printed.
- 34. Implement a function to check if a binary tree is balanced. A balanced tree is defined to be a tree in which the heights of two subtrees of any node never differ by more than one.
- 35. Draw a diagram showing how the binary tree below appears in memory when stored without pointers using a block of contiguous memory cells as described in Section 8.3.



- 36. Suppose each node in a binary tree contains a value. Design a method that prints all the paths which sum to a specified value. The path can start at an arbitrary node.
- 37. Give an example in which you might want to implement a list (the conceptual structure) as a tree (the actual underlying structure). Give an example in which you might want to implement a tree (the conceptual structure) as a list (the actual underlying structure).
- 38. Suppose a binary tree T1 has millions of nodes, and another tree T2, has hundreds of nodes. Create an algorithm to decide whether T2 is a subtree of T1. T2 would be a subtree of T1 if there is a node "n" in T1 whose subtree is identical to T2.
- 39. Design an algorithm to check whether a route between two nodes exists in a directed graph.

40. Identify the trees below whose nodes would be printed in alphabetical order by the algorithm in Figure 8.24.



41. Modify the function in Figure 8.24 to print the "list" in reverse order.
42. Suppose a call center has three levels of employees—respondent, manager, and director. An incoming telephone call must first be allocated to a respondent who is free. If the respondent cannot handle the call, the call must be escalated to a manager. If the manager is occupied, then the call should be escalated to a director. Design the classes and data structures for this problem. Implement a method `dispatchCall()` which assigns a call to the first available employee.
43. Design a procedure for finding and deleting a given value from a tree stored in the fashion of Figure 8.20.
44. In the traditional implementation of a tree, each node is constructed with a separate pointer for each possible child. The number of such pointers is a design decision and represents the maximum number of children any node can have. If a node has fewer children than pointers, some of its pointers are simply set to null. But such a node can never have more children than pointers. Describe how a tree could be implemented without limiting the number of children a node could have.
45. Using pseudocode modeled on the C struct statement introduced in Section 8.5, define a user-defined data type representing data regarding an employee of a company (such as name, address, job assignment, pay scale, and so on).
46. Using pseudocode similar to the Java class syntax of Figure 8.27, sketch a definition of an abstract data type representing a list of names. In particular, what structure would contain the list and what functions would be provided to manipulate the list? (You do not

need to include detailed descriptions of the functions.)

47. Using pseudocode similar to the Java class syntax of Figure 8.27, sketch a definition of an abstract data type representing a queue. Then give pseudocode statements showing how instances of that type could be created and how entries could be inserted in and deleted from those instances.
48. a. What is the difference between a user-defined data type and a primitive data type?
b. What is the difference between an abstract data type and a user-defined data type?
49. Identify the data structures and procedures that might appear in an abstract data type representing an address book.
50. Identify the data structures and procedures that might appear in an abstract data type representing a simple spacecraft in a video game.
51. Modify Figure 8.27 and the `StackType` interface from Section 8.5 so that the class defines a queue rather than a stack.
52. In what way is a class more general than a traditional abstract data type?
- *53. Using instructions of the form `DR0S` and `ER0S` as described at the end of Section 8.7, write a complete machine language routine to push an entry onto a stack implemented as shown in Figure 8.12. Assume that the stack pointer is in register `F` and that the entry to be pushed is in register `5`.
- *54. Suppose a Boolean expression consisting of the symbols `0`, `1`, `&`, `|`, and `A`, and a desired Boolean result value is given. Implement a function to count the number of ways to parenthesize the expression, such that it evaluates to the result. For example, if the expression is `1 A 01011` and the desired result is false (`0`), the output will be `1A( (010) 11)` and `1 A (91 (011))`.
- *55. What advantages does an instruction of the form `DR0S` as described in Section 8.7 have over an instruction of the form `DRXY`? What advantage does the form `DRXS` as described in question 4 of Section 8.7 have over the form `DR0S`?

Chapter Review Problems

(Asterisked problems are associated with optional sections.)

1. What is the significance of database management systems?
2. What is a database model?
3. What is a lossless decomposition or a nonloss decomposition?
4. What is the difference between a tuple and an attribute?
5. Identify two benefits of separating application software from the DBMS.
6. Describe the similarities between an abstract data type (Chapter 8) and a database model.
7. Identify the level within a database system (user, programmer of application software, designer of the DBMS software) at which each of the following concerns or activities occur:
 - a. How should data be stored on a disk to maximize efficiency?
 - b. Is there a vacancy on flight 243?
 - c. How should a relation be organized in mass storage?
 - d. How many times should a user be allowed to mistype a password before the conversation is terminated?
 - e. How can the **PROJECT** operation be implemented?
8. Which of the following tasks are handled by a DBMS?
 - a. Ensure that a user's access to the database is restricted to the appropriate subschema.
 - b. Translate commands stated in terms of the database model into actions compatible with the actual data storage system.
 - c. Disguise the fact that the data in the database is actually scattered among many computers in a network.
9. Describe how the following information about railways, trains (for a particular day), and passengers would be represented in a relational database:

Trains: Rolar, Omni, and Holiday

Trains for Rolar: R221, R567, and R234

Holiday special Trains: H897 and H008

Trains for Omni: O999 and O815

John has reservations on R221 (seat 34U), R567 (seat 23U), and R234 (seat 43L).

Henry has reservations on O999 (seat 15L) and H008 (seat 18L).

Duke has reservations on H897 (seat 7U) and O815 (seat 2L).

10. To what extent is the order in which **SELECT** and **PROJECT** operations are applied to a relation significant? That is, under what conditions will **SELECT**ing and then **PROJECT**ing produce the same results as first **PROJECT**ing and then **SELECT**ing?
11. Give an argument showing that the "where" clause in the **JOIN** operation as described in Section 9.2 is not necessary. (That is, show that any query that uses a "where" clause could be restated using a **JOIN** operation that concatenated every tuple in one relation with every tuple in the other.)
12. In terms of the relations shown below, what is the appearance of the relation **FINAL** after executing each of these instructions:

A relation			B relation	
L	M	N	X	Y
X	P	1	9	7
Y	Q	7	4	3
Z	R	6		

- a. **FINAL** $\leftarrow$ **SELECT** from B where X = 9
- b. **FINAL** $\leftarrow$ **PROJECT** L from A
- c. **FINAL** $\leftarrow$ **SELECT** from A where L = Z
- d. **FINAL** $\leftarrow$ **JOIN** A and B where A.N = B.Y

13. Using the commands SELECT, PROJECT, and JOIN, write a sequence of instructions to answer each of the following questions about branches and their courses in terms of the following database:

Course relation	
CName	ID
Networks	IT655
DataBase	CS543
VLSI	EC653

Branch relation		
BName	ID	Credits
Computer Science	CS543	4
Computer Science	EC653	5
Electronics & Communication	IT655	4
Electronics & Communication	EC653	5
Information Technology	CS543	4
Information Technology	IT655	4

- a. Which branches offer IT665?
 - b. List all the branches present in Branch relation.
 - c. Which branches offer 4-credit courses?
14. Answer question 13 using SQL.
15. Using the commands SELECT, PROJECT, and JOIN, write sequences to answer the following questions about the information in the EMPLOYEE, JOB, and ASSIGNMENT relations in Figure 9.5:
- a. Obtain the name of an employee whose Job ID is S25X.
 - b. Obtain a list of the department, skill code and job title of the employee named G. Jerry Smith.
 - c. Obtain a list of the name and address of the employee whose start date is 5-1-2010.
16. Answer question 14 using SQL.
17. Design a relational database containing information about authors, their titles, and their publishers. (Avoid redundancies similar to those in Figure 9.4.)
18. Design a relational database containing information about students, their courses, and their current semester along with their

disciplines. (Avoid redundancies similar to those in Figure 9.4.)

19. Design a relational database containing information about employees of different companies and their salaries. (Avoid redundancies similar to those in Figure 9.4.)
20. Design a relational database containing information about fruits, colors, and the season of their availability. (Avoid redundancies similar to those in Figure 9.4.)
21. Design a relational database containing information about parts, suppliers, and customers. Each part might be supplied by several suppliers and ordered by many customers. Each supplier might supply many parts and have many customers. Each customer might order many parts from many suppliers; in fact, the same part might be ordered from more than one supplier. (Avoid redundancies similar to those in Figure 9.4.)
22. Write a sequence of instructions (using the operations SELECT, PROJECT and JOIN) to retrieve the JobId, StartDate, and TermDate for each job in the accounting department from the relational database described in Figure 9.5.
23. Answer the previous problem using SQL.
24. Write a sequence of instructions (using the operations SELECT, PROJECT and JOIN) to retrieve the Name, Address, JobTitle, and Dept of every current employee from the relational database described in Figure 9.5.
25. Answer the previous problem using SQL.
26. Write a sequence of instructions (using the operations SELECT, PROJECT and JOIN) to retrieve the Name and JobTitle of each current employee from the relational database described in Figure 9.5.
27. Answer the previous problem using SQL.
28. What is the difference in the information supplied by the single relation

Name	Department	TelephoneNumber
Jones	Sales	555-2222
Smith	Sales	555-3333
Baker	Personnel	555-4444

and the two relations

Name	Department
Jones	Sales
Smith	Sales
Baker	Personnel

Department	TelephoneNumber
Sales	555-2222
Sales	555-3333
Personnel	555-4444

29. Design a relational database containing information about novels, its authors and the number of pages in the novels. Be sure to allow for the fact that a novel may contain less number of pages in one version and at the same it may contain more number of pages in a new version.

30. Pick a popular website such as www.google.com, www.amazon.com, or www.ebay.com and design a relational database that you would propose to serve as the site's supporting database.

31. On the basis of the database represented in Figure 9.5, state the question that is answered by the following program segment:

```
TEMP ← SELECT from JOB
      where JobTitle = 'Secretary'
RESULT ← PROJECT SkillCode, Department
         from TEMP
```

32. Answer question 31 using SQL.

33. On the basis of the database represented in Figure 9.5, state the question that is answered by the following program segment:

```
INTER1 ← JOIN JOB and ASSIGNMENT
         where JOB.JobId =
               ASSIGNMENT.JobId
INTER2 ← SELECT from INTER1 where
         EmpId = '23Y34'
FINAL ← PROJECT TermDate, StartDate
         from INTER2
```

34. Answer question 33 using SQL.

35. On the basis of the database represented in Figure 9.5, state the question that is answered by the following program segment:

```
TEMP1 ← JOIN EMPLOYEE and JOB
         where EMPLOYEE.EmpId = JOB.EmpId
```

36. Translate the query in the previous problem into SQL.

37. Translate the SQL statement

```
SELECT Job.JobTitle
FROM Assignment, Job
WHERE Assignment.JobId = Job.JobId
      AND Assignment.EmpId = '34Y70';
```

into a sequence of **SELECT**, **PROJECT**, and **JOIN** operations.

38. Translate the SQL statement

```
SELECT Assignment.StartDate
FROM Assignment, Employee
WHERE Assignment.EmpId =
      Employee.EmpId
      AND Employee.Name = 'Joe E.
      Baker';
```

into a sequence of **SELECT**, **PROJECT**, and **JOIN** operations.

39. Describe the effect that the following SQL statement would have on the database in Problem 13.

```
INSERT INTO Course
VALUES ('Computer Concepts', 'CS342');
```

40. Describe the effect that the following SQL statement would have on the database in question 13.

```
UPDATE Branch
SET ID = 'CS555'
WHERE BName = 'Computer Science'
      AND Credits = 5;
```

- *41. Identify some of the objects that you would expect to find in an object-oriented database used to maintain a grocery store's inventory. What methods would you expect to find within each of these objects?

- *42. Identify some of the objects that you would expect to find in an object-oriented database used to maintain records of a library's holdings. What methods would you expect to find within each of these objects?

- *43. What incorrect information is generated by the following schedule of transactions T1 and T2?

- T1 is designed to compute the sum of accounts A and B; T2 is designed to transfer \$100 from account A to account B. T1 begins by retrieving the balance of account A; then, T2 performs its transfer; and finally, T1 retrieves the balance of account B and reports the sum of the values it has retrieved.
- *44. What is the difference between incorrect summary problem and lost update problem? Explain your answer with examples.
 - *45. What effect would the wound-wait protocol have on the sequence of events in question 43 if T1 was the younger transaction? If T2 was the younger transaction?
 - *46. Suppose one transaction tries to add \$100 to an account whose balance is \$200 while another tries to withdraw \$100 from the same account. Describe an interweaving of these transactions that would lead to a final balance of \$100. Describe an interweaving of these transactions that would lead to a final balance of \$300.
 - *47. What is the difference between a transaction having exclusive access or shared access to an item in a database and why is the distinction important?
 - *48. The problems discussed in Section 9.4 involving concurrent transactions are not limited to database environments. What similar problems would arise when accessing a document with word processors? (If you have a PC with a word processor, try to access the same document with two activations of the word processor and see what happens.)
 - *49. Suppose a sequential file contains 50,000 records and 5 milliseconds is required to interrogate an entry. How long should we expect to wait when retrieving a record from the middle of the file?
 - *50. List the steps that are executed in the merge algorithm in Figure 9.15 if one of the input files is empty at the start.
 - *51. Modify the algorithm in Figure 9.15 to handle the case in which both input files contain a record with the same key field value. Assume that these records are identical and that only one should appear in the output file.
 - *52. Design a system by which a file stored on a disk can be processed as a sequential file with either of two different orderings.
 - *53. Describe how a sequential file containing information about a magazine's subscribers could be constructed using a text file as the underlying structure.
 - *54. Design a technique by which a sequential file whose logical records are not a consistent size could be implemented as a text file. For example, suppose you wanted to construct a sequential file in which each logical record contained information about a novelist as well as a list of that author's works.
 - *55. Explain the terms cluster analysis, association analysis, outlier analysis, and sequential pattern analysis in brief.
 - *56. The chapter drew parallels between a traditional file index and the file directory system maintained by an operating system. In what ways does an operating system's file directory differ from a traditional index?
 - *57. If a hash file is partitioned into 10 buckets, what is the probability of at least two of three arbitrary records hashing to the same bucket? (Assume the hash function gives no bucket priority over the others.) How many records must be stored in the file until it is more likely for collisions to occur than not?
 - *58. Solve the previous problem, assuming that the file is partitioned into 100 buckets instead of 10.
 - *59. If we are using the division technique discussed in this chapter as a hash function and the file storage area is divided into 23 buckets, which section should we search to find the record whose key, when interpreted as a binary value, is the integer 124?
 - *60. Compare the implementation of a hash file to that of a homogeneous two-dimensional array. How are the roles of the hash function and the address polynomial similar?
 - *61. Give one advantage that
 - a. a sequential file has over an indexed file.
 - b. a sequential file has over a hash file.
 - c. an indexed file has over a sequential file.
 - d. an indexed file has over a hash file.
 - e. a hash file has over a sequential file.
 - f. a hash file has over an indexed file.
 - *62. Explain the technique of hashing. How is it related to a hash function and a hash table?

Chapter Review Problems

1. Show how a structure of the form

```
while X equals 0:
```

```
·  
·  
·
```

can be simulated with Bare Bones.

2. Write a Bare Bones program that places a 1 in the variable Z if the variable X is less than or equal to the variable Y, and places a 0 in the variable Z otherwise.
3. Write a Bare Bones program that places the Xth power of 2 in the variable Z.

```
if (the Nth List entry is less  
    than the entry preceding it):  
    interchange the Nth entry with  
        the preceding entry  
N = N - 1
```

48. Use RSA public-key encryption to encrypt the message 1111 using the public keys $n = 119$ and $e = 5$.
49. Use RSA public-key encryption to decrypt the message 1101 using the private keys $n = 911$ and $d = 5$.
50. Suppose you know that the public keys to a public-key encryption system based on the

4. In each of the following cases write a program sequence in Bare Bones that performs the indicated activity:
- Assign 0 to Z if the value of X is even; otherwise assign 1 to Z.
 - Calculate the sum of the integers from 0 to X.
5. Write a Bare Bones routine that divides the value of X by the value of Y. Disregard any remainder; that is, 1 divided by 2 produces 0, and 5 divided by 3 produces 1.
6. Describe the function computed by the following Bare Bones program, assuming the

RSA algorithm are $n = 3233$ and $e = 17$. What are the private keys? Can you solve this problem in a reasonable amount of time?

51. Find the factors of 211,397. How does this problem relate to this chapter?
52. What can be concluded if the positive integer n has no integer factors in the range from 2 to the square root of n ? What does this tell you about the task of finding the factors of a positive integer?

function's inputs are represented by X and Y and its output by Z:

```
copy X to Z
copy Y to Aux
while Aux not 0:
    decr Z
    decr Aux
```

7. Describe the function computed by the following Bare Bones program, assuming the function's inputs are represented by X and Y and its output by Z:

```
clear Z
copy X to Aux1
copy Y to Aux2
while Aux1 not 0:
    while Aux2 not 0:
        decr Z
        decr Aux2
    decr Aux1
```

8. Write a Bare Bones program that computes the exclusive-OR of the variables X and Y, leaving the result in the variable Z. You might assume that X and Y start only with integer values of 0 and 1.
9. Show that if we allow instructions in a Bare Bones program to be labeled with integer values and replace the `while` loop structure with the conditional branch represented by the form
- ```
if name not 0:
 goto label
```

where `name` is any variable and `label` is an integer value used elsewhere to label an instruction, then the new language will still be a universal programming language.

10. In this chapter we saw how the statement

```
copy name1 to name2
```

could be simulated in Bare Bones. Show how that statement could still be simulated if the `while` loop structure in Bare Bones were replaced with a posttest loop expressed in the form

```
repeat:...
 until (name equals 0)
```

11. Show that the Bare Bones language would remain a universal language if the `while`

statement were replaced with a posttest loop expressed in the form

```
repeat:
 ...
 until (name equals 0)
```

12. Give an example of a language that is not deterministic.
13. Design a Turing machine that recognizes the set of palindromes in  $\{a, b\}^*$  such that the number of `a`s is a multiple of three.
14. Suppose a pattern of 0s and 1s on the tape of a Turing machine is delimited by asterisks at either end. Design a Turing machine that rotates this pattern one cell to the left, assuming that the machine starts with the current cell being the asterisk at the right end of the pattern.
15. Design a Turing machine that places 0s and 1s in all the cells to the right of the current cell until it reaches a cell containing a plus symbol.
16. Summarize the Church–Turing thesis.
17. Is the following Bare Bones program self-terminating? Explain your answer.

```
copy X to Y
incr Y
incr Y
while X not 0:
 decr X
 decr X
 decr Y
 decr Y
decr Y
while Y not 0:
 incr X
 decr Y
while X not 0:
```

18. Is the following Bare Bones program self-terminating? Explain your answer.
- ```
while X not 0:
```
19. Is the following Bare Bones program self-terminating? Explain your answer.

```
while X not 0:
    decr X
```

20. Analyze the validity of the following pair of statements:
The next statement is true.
The previous statement is false.
21. Analyze the validity of the statement "The cook on a ship cooks for all those and only those who do not cook for themselves." (Who cooks for the cook?)
22. Suppose you were in a country where each person was either a truth teller or a liar. (A truth teller always tells the truth, a liar always lies.) What single question could you ask a person that would allow you to detect whether that person was a truth teller or a liar?
23. When is a function said to be Turing computable? Explain your answer.
24. What is meant by a halting problem? Explain how a halting problem is undecidable.
25. Suppose you needed to find out if anyone in a group of people had a birthday on a particular date. One approach would be to ask the members one at a time. If you took this approach, the occurrence of what event would tell you that there was such a person? What event would tell you that there was no such person? Now suppose that you wanted to find out if at least one of the positive integers has a particular property and you applied the same approach of systematically testing the integers one at a time. If, in fact, some integer has the property, how would you find out? If, however, no integer has the property, how would you find out? Is the task of testing to see if a conjecture is true necessarily symmetric with the task of testing to see if it is false?
26. Is the problem of searching through a binary tree for a particular value a polynomial problem? Justify your answer.
27. Design an algorithm for calculating the Fibonacci series of a given number. Is your solution efficient? Is your solution a polynomial or nonpolynomial one?
28. Give an example of a scenario in which there is a fine line of distinction between deterministic and nondeterministic algorithms.
29. Design a Turing machine for a palindrome detector. The program should check a string of symbols on the tape and return a `true(1)` if it is a palindrome, else `false(0)`.
30. Charlie Programmer is given the problem of dividing a group (of an even number of people) into two disjoint subgroups of equal size so that the difference between the total ages of each subgroup is as large as possible. He proposes the solution of forming all possible subgroup pairs, computing the difference between the age totals of each pair, and selecting the pair with the largest difference. Mary Programmer, on the other hand, proposes that the original group first be sorted by age and then divided into two subgroups by forming one subgroup from the younger half of the sorted group and the other from the older half. What is the complexity of each of these solutions? Is the problem itself of polynomial, NP, or nonpolynomial complexity?
31. Prove that if L is recognized by a Turing machine with a two-way infinite tape, it can be recognized by a Turing machine with a one-way infinite tape.
32. Suppose a lottery is based on correctly picking four integer values, each in the range from 1 to 50. Moreover, suppose that the jackpot grows so large that it becomes profitable to buy a separate lottery ticket for each possible combination. If it takes one second to buy a single ticket, how long would it take to buy one ticket for each combination? How would the time requirement change if the lottery required picking five numbers instead of four? What does this problem have to do with the material from this chapter?
33. Is the following algorithm deterministic? Explain your answer.

```
def mystery(Number):  
    if (Number > 5):  
        answer "yes"  
    else:  
        pick a value less than 5 and give  
        this number as the answer
```


34. Is the following algorithm deterministic? Explain your answer.
- ```
Drive straight ahead.
At the third intersection, ask the
 person standing on the corner if you
 should turn right or left.
Turn according to that person's
 directions.
Drive two more blocks and stop
 there.
```

35. Identify the points of nondeterminism in the following algorithm:
- ```
Select three numbers between 1 and 100.  
if (the sum of the selected  
  numbers is greater than 150):  
  answer "yes"  
else:  
  select one of the chosen numbers  
  and give that number as the  
  answer
```

36. Does the following algorithm have a polynomial or nonpolynomial time complexity? Explain your answer.

```
def mystery(ListOfNumbers):  
    Pick a collection of numbers from  
    ListOfNumbers.  
    if (the numbers in that collection  
        add to 125):  
        answer "yes"  
    else:  
        do not give an answer
```

37. Which of the following problems are in the class P?
- For a given set S of n integers, sort S .
 - The traveling salesperson problem.
 - The Hamilton Path problem.
 - The Node Cover problem.
38. Summarize the distinction between stating that a problem is a polynomial problem and stating that it is a nondeterministic polynomial problem.
39. Explain how a Turing machine can be used to determine whether a given number is prime.
40. Suppose you are given two algorithms for solving the same problem. One algorithm has time complexity n^4 and the other has time

complexity $4n$. For what size inputs is the former more efficient than the latter?

41. Suppose we were faced with solving the traveling salesperson problem in a context involving 15 cities in which any two cities were connected by a unique road. How many different paths through the cities would there be? How long would it take to compute the length of all of these paths assuming that the length of a path can be computed in one microsecond?
42. How many comparisons between names are made if the merge sort algorithm (Figures 12.9 and 12.8) is applied to the list Alice, Bob, Carol, and David? How many are required if the list is Alice, Bob, Carol, David, and Elaine?
43. How does a universal Turing machine function? Explain with an example.
44. Design an algorithm for finding integer solutions for equations of the form $x^2 + y^2 = n$, where n is some given positive integer. Determine the time complexity of your algorithm.
45. Another problem that falls in the NP-complete category is the **knapsack problem**, which is the problem of finding which numbers from a list are the ones whose sum is a particular value. For example, the numbers 257, 388, and 782 are the entries in the list

642 257 771 388 391 782 304

whose sum is 1427. Find the entries whose sum is 1723. What algorithm did you apply? What is the complexity of that algorithm?

46. Identify similarities between the traveling salesperson problem and the knapsack problem (see question 45).
47. The following algorithm for sorting a list is called the bubble sort. How many comparisons between list entries does the bubble sort require when applied to a list of n entries?

```
def BubbleSort(List):  
    Counter = 1  
    while (Counter < number of entries  
        in List):  
        N = the number of entries in List  
        while (N > 1):
```